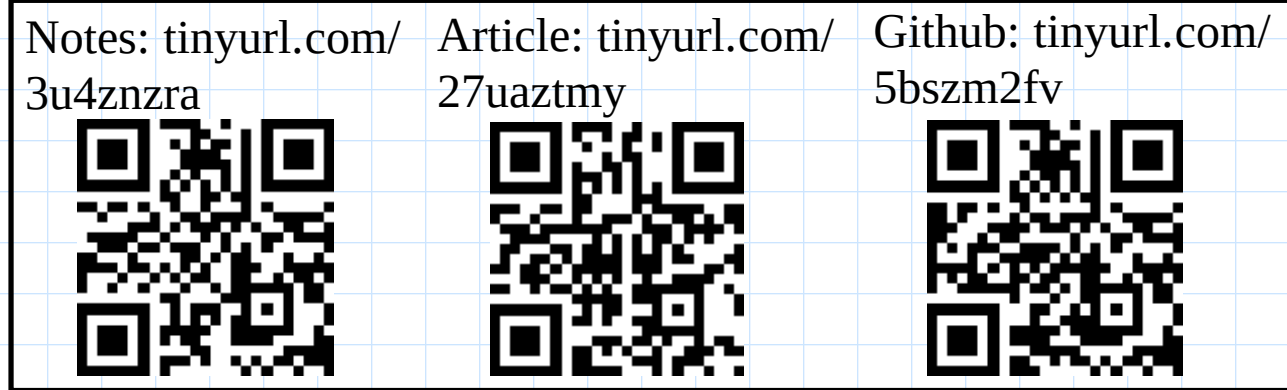
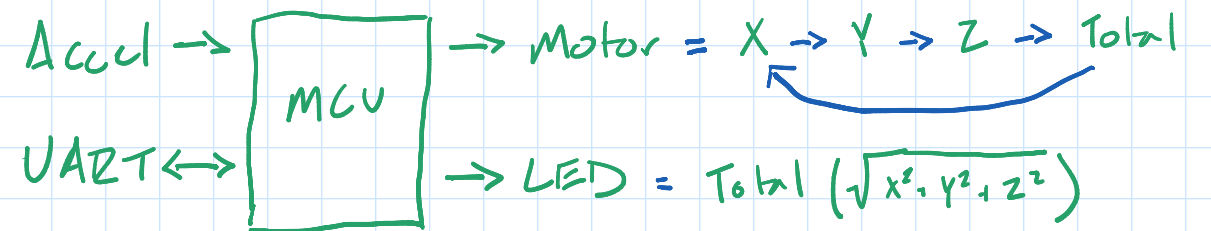


Enter the Matrix: Simulating your Embedded System



An example system:



```

void main(void) {
    while(1) {
        ...
        HAL_GPIO_WritePin(LED_GPIO_Port, LED_Pin, GPIO_PIN_SET);
        ...
    }
}
    
```

❌ ① Isolate the HW

#include "hardware.h"

```

void main(void) {
    while(1) {
        ...
        turnOnLed();
        ...
    }
}
    
```

Only one given to linker

```

// stm32.c
void turnOnLed(void) {
    HAL_GPIO_WritePin(...);
}

// x86.c
void turnOnLed(void) {
    printf("LED is on\n");
}
    
```

Benefits

① Start development ASAP (no HW req'd!)

② No HW bottlenecks

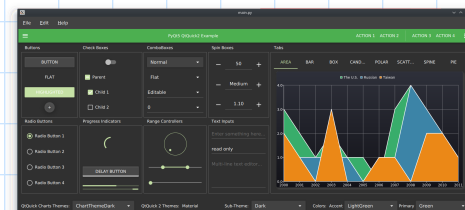
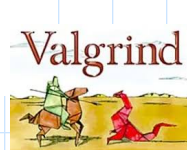
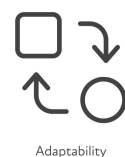
③ Isolate faults to HW

④ Easily adapt to project changes

⑤ Access to host-only programs/features:

- Valgrind
- GDB reverse-debugging

⑥ Convert simulator into dashboard later on



```

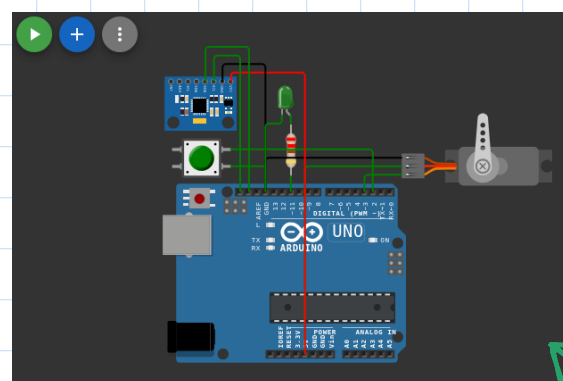
Setting motor speed: 0.100000
Setting LED brightness: 0.100000
x 2
Setting motor speed: 0.200000
Setting LED brightness: 0.200000
Setting motor speed: 0.200000
Setting LED brightness: 0.200000
    
```

```

+-----+
|Values generated: Randomly|
|Task period: 10 ms|
|Volatility: 0.200000|
+-----+
|X: -12.635740+---|Exp-
+-----+ |x a
|Y: -6.621343+---|
+-----+ |w:
    
```

```

printf("\033\033[;H");
printf(background, rand
led val, curr_x, spe
printf("\0338");
    
```



Init:

```

// Create widgets
// Create layout
// Add widgets to layout
// Register cbks
    
```

Tap btn pressed:

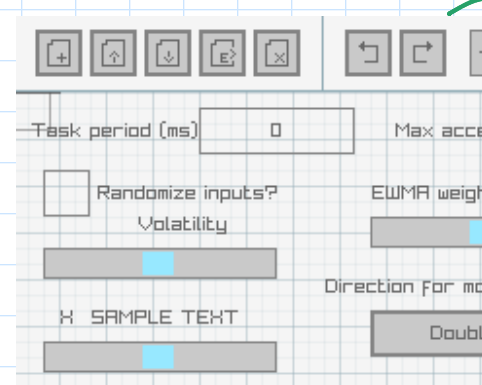
```

// Send "t" to app
    
```

Every 10 ms:

```

// Read serial
// If "m VAL":
// Append VAL to motor
// speed graph
    
```



Console

scanf
read → App → printf
fgetc write
Use 2nd thread
for blocking functions

char background[] =

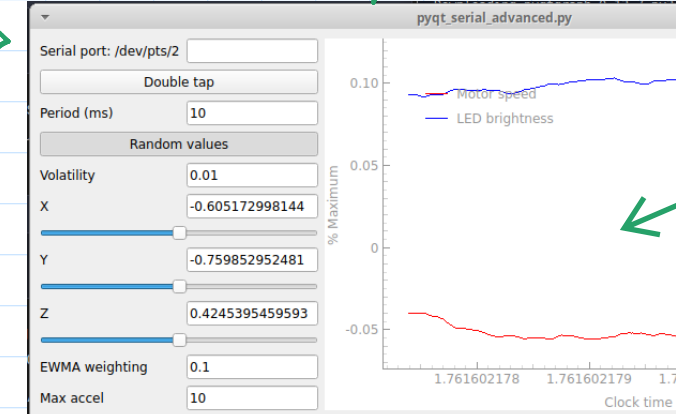
```

+-----+
|Values generated: %.8s" |
|Task period: %8d" ms |
|Volatility: %8lf" |
+-----+
|X: %12lf" +--->|Ex
+-----+ |X
|Y: %12lf" +--->|
+-----+
    
```

Virtual Hardware
e.g. Qemu Renode
TinkerCAD
Wokwi

GUIs !! WARNING: High complexity ahead

PyQT Retained-Mode



```

void main(void) {
    while(1) {
        // Make and handle widgets, e.g.
        if( makeButton() ) buttonWasPressed();

        runTheApp();
    }
}
    
```

Immediate-mode 😊

```

***** x86.c *****
double curr_x;

void initHardware(void) {
    pthread_create(...)
}

void* scanfPthread(void* data) {
    while(1) {
        scanf(...)
        // "x val"? curr_x = val;
    }
}

void readAccel_gs(double* x) {
    *x = curr_x;
}

void setMotorSpeed(double speed) {
    printf("%lf\n", speed);
}
    
```

```

***** app.c *****
void runTheApp(void) {
    // NO WHILE LOOP
    readAccel(&x, &y, &z);
}
    
```

```

***** wokwi.ino *****
void setup() {
    // Initialize HW
}

void loop() {
    runTheApp();
}
    
```

```

***** x86.c *****
void* readThread(void* data) {
    while(1) {
        scanf(...)
        // Recv'd "t"? tapCallback();
    }
}

void setMotorSpeed(double speed) {
    // Write "m <speed>\n"
}
    
```

raygui

